



Traffic Mirroring

Amazon Virtual Private Cloud



Amazon Virtual Private Cloud: Traffic Mirroring

Copyright © 2025 Amazon Web Services, Inc. and/or its affiliates. All rights reserved.

Amazon's trademarks and trade dress may not be used in connection with any product or service that is not Amazon's, in any manner that is likely to cause confusion among customers, or in any manner that disparages or discredits Amazon. All other trademarks not owned by Amazon are the property of their respective owners, who may or may not be affiliated with, connected to, or sponsored by Amazon.

Table of Contents

What is Traffic Mirroring?	1
Traffic Mirroring concepts	1
Work with Traffic Mirroring	1
Traffic Mirroring benefits	2
Regional availability	2
Supported instance types	2
Virtualized instances	2
Bare metal instances	3
Pricing	3
How Traffic Mirroring works	5
Targets	6
Network interfaces	7
Network Load Balancer	7
Gateway Load Balancer endpoints	8
VXLAN encapsulation	9
Routing and security groups	9
Processing mirrored traffic	10
Filters	10
Sessions	11
Traffic mirror sources	11
Connectivity options	12
Packet format	13
Get started	15
Prerequisites	16
Step 1: Create the traffic mirror target	16
Step 2: Create the traffic mirror filter	16
Step 3: Create the traffic mirror session	17
Step 4: Analyze the data	18
Traffic Mirroring example scenarios	19
Mirror inbound TCP traffic to a single appliance	19
Step 1: Create a traffic mirror target	20
Step 2: Create a traffic mirror filter	20
Step 3: Create a traffic mirror session	21
Mirror inbound TCP and UDP traffic to multiple appliances	21

Step 1: Create a traffic mirror target for Appliance A	23
Step 2: Create a traffic mirror target for Appliance B	23
Step 3: Create a traffic mirror filter with a rule for TCP traffic	23
Step 4: Create a traffic mirror filter with a rule for UDP traffic	23
Step 5: Create a traffic mirror session for the TCP traffic	24
Step 6: Create a traffic mirror session for the UDP traffic	25
Mirror non-local VPC traffic	25
Step 1: Create a traffic mirror target	26
Step 2: Create a traffic mirror filter	26
Step 3: Create a traffic mirror session	29
Mirror traffic to a Gateway Load Balancer endpoint	29
Step 1: Create a traffic mirror target in Spoke VPC1	31
Step 2: Create a traffic mirror target in Spoke VPC2	31
Step 3: Create a traffic mirror filter rule	31
Step 4: Create a traffic mirror session in Spoke VPC1	32
Step 5: Create a traffic mirror session in Spoke VPC2	32
Work with Traffic Mirroring	33
Create or delete a traffic mirror target	33
View traffic mirror targets and modify target tags	35
Share a traffic mirror target	35
Accept or delete a shared traffic mirror target	36
Create, modify, or delete a traffic mirror filter	37
Create, modify, or delete a traffic mirror session	39
Work with open-source tools	41
Step 1: Install the Suricata software on the EC2 instance target	41
Step 2: Create a traffic mirror target	42
Step 3: Create a traffic mirror filter	42
Step 4: Create a traffic mirror session	43
Monitor mirrored traffic	44
Traffic Mirroring metrics and dimensions	44
View Traffic Mirroring CloudWatch metrics	47
Limitations	48
General limitations	48
MTU and packet length limitations	49
Traffic bandwidth and prioritization limitations	49
Checksum offloading limitations	50

Quotas	51
Sessions	51
Targets	51
Filters	52
Throughput	52
Packets	52
Sources	52
Identity and access management	55
.....	55
Document history	57

What is Traffic Mirroring?

Traffic Mirroring is an Amazon VPC feature that you can use to copy network traffic from an elastic network interface of type `interface`. You can then send the traffic to out-of-band security and monitoring appliances for:

- Content inspection
- Threat monitoring
- Troubleshooting

The security and monitoring appliances can be deployed as individual instances, or as a fleet of instances behind either a Network Load Balancer or a Gateway Load Balancer with a UDP listener. Traffic Mirroring supports filters and packet truncation, so that you can extract only the traffic of interest, using the monitoring tools of your choice.

Traffic Mirroring concepts

The following are the key concepts for Traffic Mirroring:

- **Source** — The network interface to monitor.
- **Filter** — A set of rules that defines the traffic that is mirrored.
- **Target** — The destination for mirrored traffic.
- **Session** — Establishes a relationship between a source, a filter, and a target.

Work with Traffic Mirroring

You can create, access, and manage your traffic mirror resources using any of the following:

- **AWS Management Console**— Provides a web interface that you can use to access your traffic mirror resources.
- **AWS Command Line Interface (AWS CLI)** — Provides commands for a broad set of AWS services, including Amazon VPC. The AWS CLI is supported on Windows, macOS, and Linux. For more information, see [AWS Command Line Interface](#).

- **AWS SDKs** — Provide language-specific APIs. The AWS SDKs take care of many of the connection details, such as calculating signatures, handling request retries, and handling errors. For more information, see [AWS SDKs](#).
- **Query API**— Provides low-level API actions that you call using HTTPS requests. Using the Query API is the most direct way to access Amazon VPC. However, it requires that your application handle low-level details such as generating the hash to sign the request and handling errors. For more information, see [Amazon VPC actions](#) in the *Amazon EC2 API Reference*.

Traffic Mirroring benefits

Traffic Mirroring offers the following benefits:

- **Simplified operation** — Mirror any range of your VPC traffic without having to manage packet forwarding agents on your EC2 instances.
- **Enhanced security** — Capture packets at the elastic network interface, which cannot be disabled or tampered with from a user space.
- **Increased monitoring options** — Send your mirrored traffic to any security device.

Regional availability

Traffic Mirroring is available in all Regions.

Supported instance types

Instance types in the following instance families are supported as Traffic Mirroring source.

Virtualized instances

Virtualized instance types in the following instance families are supported as Traffic Mirroring source:

- **General purpose:** A1 | M4 | M5 | M5a | M5ad | M5d | M5dn | M5n | M5zn | M6a | M6g | M6gd | M6i | M6id | M6idn | M6in | M7a | M7g | M7gd | M7i | M7i-flex | Mac1 | Mac2 | Mac2-m1ultra | Mac2-m2 | Mac2-m2pro | T3 | T3a | T4g
- **Compute optimized:** C4 | C5 | C5a | C5ad | C5d | C5n | C6a | C6g | C6gd | C6gn | C6i | C6id | C6in | C7a | C7g | C7gd | C7i | C7i-flex

- **Memory optimized:** R4 | R5 | R5a | R5ad | R5b | R5d | R5dn | R5n | R6a | R6g | R6gd | R6i | R6id | R6idn | R6in | R7a | R7g | R7gd | R7i | R7iz | U-3tb1 | U-6tb1 | U-9tb1 | U-12tb1 | U-18tb1 | U-24tb1 | U7i-6tb | U7i-8tb | U7i-12tb | U7in-16tb | U7in-24tb | U7in-32tb | U7inh-32tb | X1 | X1e | X2gd | X2idn | X2iedn | X2iezn | z1d
- **Storage optimized:** D2 | D3 | D3en | H1 | I3 | I3en | I4g | I4i | I7i | Im4gn | Is4gen
- **Accelerated computing:** DL1 | DL2q | F1 | F2 | G3 | G4ad | G4dn | G5 | G5g | G6 | G6e | G6f | Gr6 | Gr6f | Inf1 | Inf2 | P3 | P3dn | P4d | P4de | P5 | P5e | Trn1 | Trn1n | VT1
- **High-performance computing:** Hpc6a | Hpc6id | Hpc7a

Bare metal instances

Bare metal instance types in the following instance families are supported as Traffic Mirroring source:

- **General purpose:** M5 | M5d | M6g | M6gd | Mac1 | Mac2 | Mac2-m1ultra | Mac2-m2 | Mac2-m2pro | A1
- **Compute optimized:** C5 | C5d | C6g | C6gd
- **Memory optimized:** R5 | R5b | R5d | R6g | R6gd | X2gd | z1d
- **Storage optimized:** I3
- **Accelerated computing:** G5g
- **Previous generation:** A1

Note

Only Nitro v2 bare-metal instances are supported as Traffic Mirroring source. Bare metal instances of any other Nitro version are not supported as Traffic Mirroring source.

Pricing

You are charged on an hourly basis for each active traffic mirror session. You'll continue to be charged for Traffic Mirroring until you [delete all active traffic mirror sessions](#). For example, you'll still be charged in the following scenarios:

- You detached the network interface from the mirror source

- You stopped or terminated the mirror source
- You changed the instance type of the mirror source to an unsupported instance type

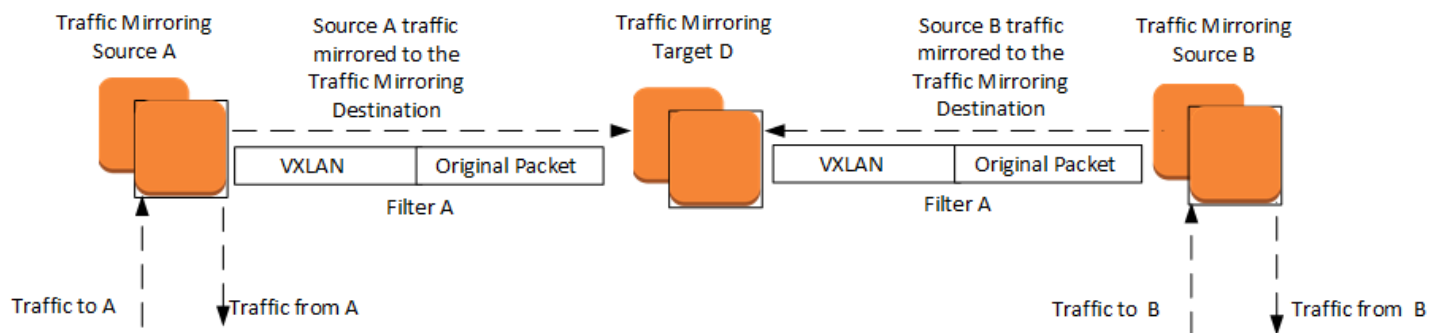
[Data transfer charges apply](#). If your traffic mirroring targets are behind a gateway or network load balancer, data processing for the load balancing services also applies.

For information about pricing for Traffic Mirroring, see **Network Analysis** on the [Amazon VPC pricing](#) page.

How Traffic Mirroring works

Traffic Mirroring copies inbound and outbound traffic from the network interfaces that are attached to your instances. You can send the mirrored traffic to the network interface of another instance, a Network Load Balancer that has a UDP listener, or a Gateway Load Balancer that has a UDP listener. The traffic mirror source and the traffic mirror target (monitoring appliance) can be in the same VPC. Or they can be in a different VPCs that are connected through intra-Region VPC peering, a transit gateway, or by a Gateway Load Balancer endpoint to connect to a Gateway Load Balancer in a different VPC.

Consider the following scenario, where you mirror traffic from two sources (Source A and Source B) to a single traffic mirror target (Target D). After you create the traffic mirror session, any traffic that matches the filter rules is encapsulated in a VXLAN header. It is then sent to the target.



The following procedures are required:

- Identify the traffic mirror source (Source A)
- Identify the traffic mirror source (Source B)
- Configure the traffic mirror target (Target D)
- Configure the traffic mirror filter (Filter A)
- Configure the traffic mirror session for Source A, Filter A, and Target D
- Configure the traffic mirror session for Source B, Filter A, and Target D

Contents

- [Understand traffic mirror target concepts](#)
- [Understand traffic mirror filter concepts](#)
- [Understand traffic mirror session concepts](#)

- [Understand traffic mirror source and target connectivity options](#)
- [Understanding traffic mirror packet format](#)

Understand traffic mirror target concepts

A *traffic mirror target* is the destination for mirrored traffic. In Amazon VPC, traffic mirroring allows you to copy network traffic from an elastic network interface (ENI) and send it to a traffic mirror target for monitoring and analysis purposes.

You can use the following resources as traffic mirror targets:

- [Network interfaces](#) of type interface
- [Network Load Balancers](#)
- [Gateway Load Balancer endpoints](#)

For high availability, we recommend that you use a Network Load Balancer or a Gateway Load Balancer endpoint as a mirror target.

You might experience out-of-order delivery of mirrored packets when you use a Network Load Balancer or Gateway Load Balancer endpoint as your traffic mirror target. If your monitoring appliance can't handle out-of-order packets, we recommend using a network interface as your traffic mirror target.

A traffic mirror target can be owned by an AWS account that is different from the traffic mirror source.

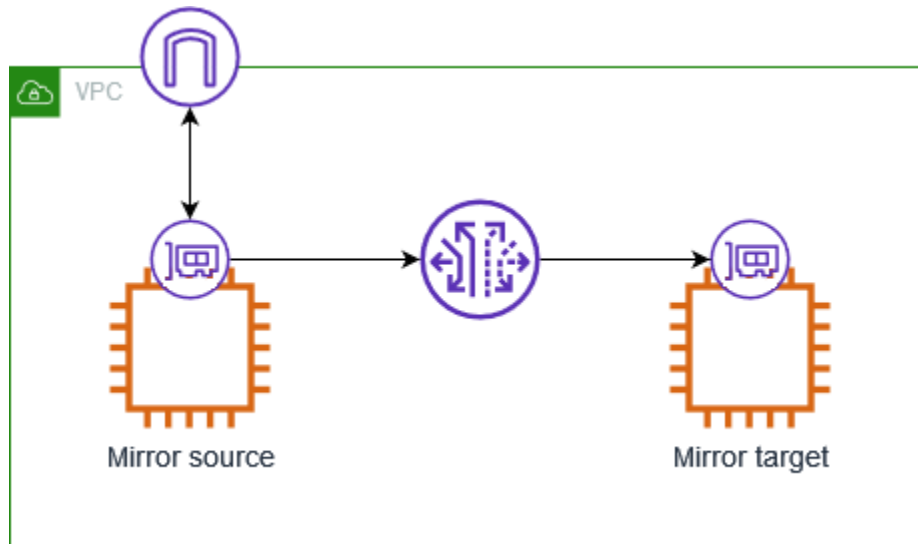
After you create a traffic mirror target, you add it to a traffic mirror session. You can use a traffic mirror target in more than one traffic mirror session. For more information, see [the section called "Sessions"](#).

Note

If the underlying resource chosen as the target is deleted, we stop traffic mirroring for any sessions that use that target. To update a traffic mirroring target and resume the traffic mirroring session, you can use [modify-traffic-mirror-session](#).

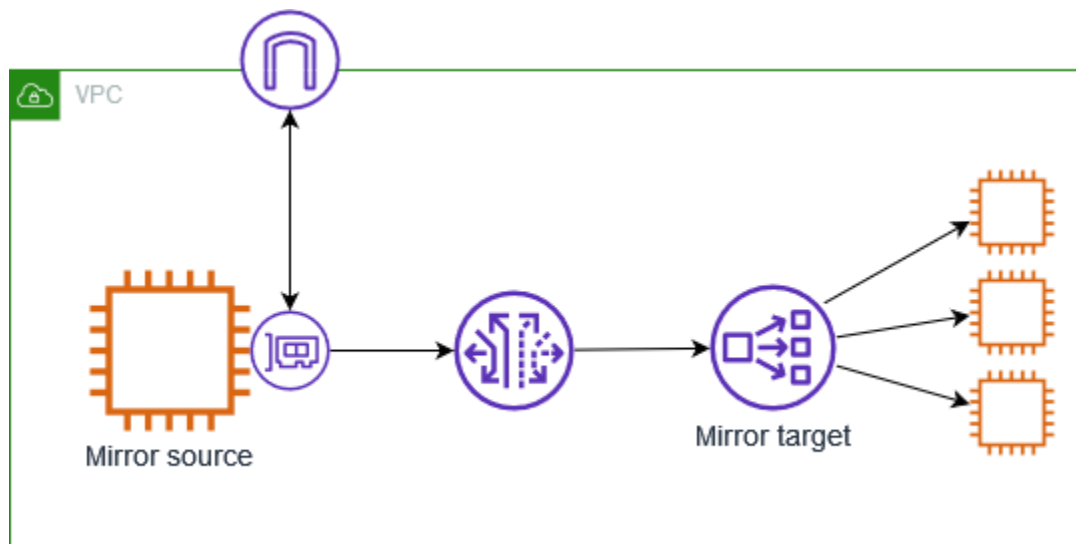
Network interfaces

The following diagram shows a traffic mirror session where the traffic mirror target is a network interface for an EC2 instance. Traffic Mirroring filters the traffic from the network interface of the mirror source and sends the accepted mirrored traffic to the mirror target.



Network Load Balancer

The following diagram shows a traffic mirror session where the traffic mirror target is a Network Load Balancer. You install the monitoring software on the target instances, and then register them with the load balancer. Traffic Mirroring filters the traffic from the network interface of the mirror source and sends the accepted mirrored traffic to the load balancer. The load balancer sends the mirrored traffic to the target instances.

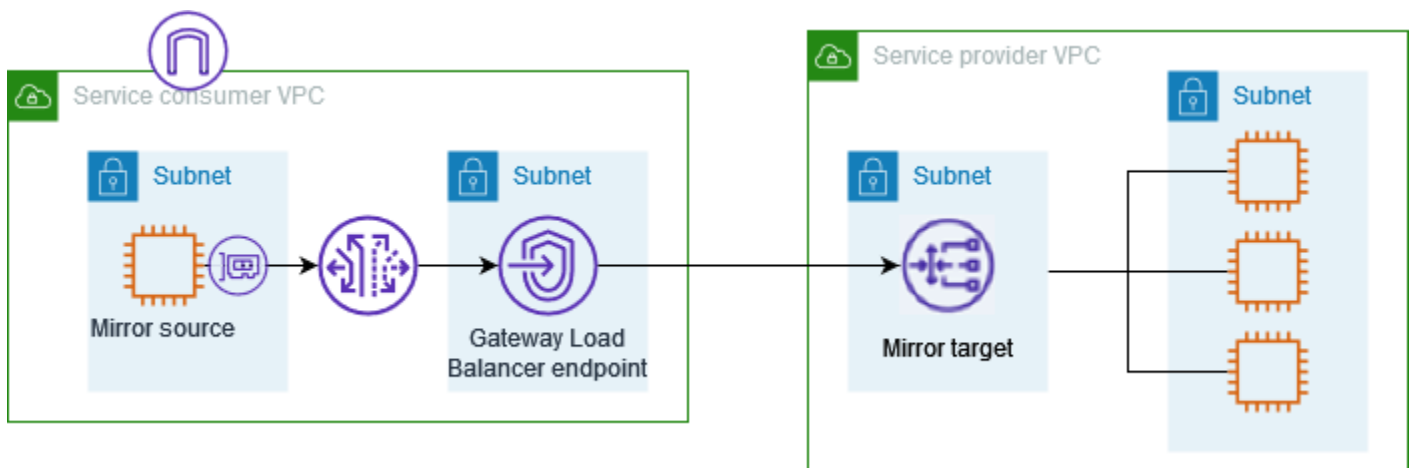


Considerations

- Traffic mirroring can't occur unless the load balancer has UDP listeners on port 4789. If you remove the UDP listeners, Traffic Mirroring fails without an error indication.
- To improve availability and fault tolerance, we recommend that you select at least two Availability Zones when you create the Network Load Balancer, and that you register target instances in each selected Availability Zone.
- We recommend that you enable cross-zone load balancing for your Network Load Balancer. If all registered target instances in an Availability Zone become unhealthy and cross-zone load balancing is disabled, the load balancer can't send the mirrored traffic to target instances in another Availability Zone. If you enable cross-zone load balancing, the load balancer can send the mirrored traffic to healthy target instances in another Availability Zone.
- If you select an additional Availability Zone for your Network Load Balancer after you create it, Traffic Mirroring does not send mirrored traffic to target instances in the new Availability Zone unless you enable cross-zone load balancing.
- When the Network Load Balancer removes a load balancer node from the DNS table, Traffic Mirroring continues to send the mirrored traffic to that node.

Gateway Load Balancer endpoints

The following diagram shows a traffic mirror session where the traffic mirror target is a Gateway Load Balancer endpoint. The mirror source is in the service consumer VPC and the Gateway Load Balancer is in the service provider VPC. You install the monitoring software on the target appliances, and then register them with the load balancer. Traffic Mirroring filters the traffic from the network interface of the mirror source and sends the accepted mirrored traffic to the Gateway Load Balancer endpoint. The load balancer sends the mirrored traffic to the target appliances.



Considerations

- A listener for Gateway Load Balancers listens for all IP packets across all ports, and then forwards traffic to the target group.
- To improve availability and fault tolerance, we recommend that you select at least two Availability Zones when you create the Gateway Load Balancer, and that you register target appliances in each selected Availability Zone.
- If all registered target appliances in an Availability Zone become unhealthy and cross-zone load balancing is disabled, the load balancer can't send the mirrored traffic to target appliances in another Availability Zone. If you enable cross-zone load balancing, the load balancer can send the mirrored traffic to healthy target appliances in another Availability Zone.
- The maximum MTU supported by the Gateway Load Balancer is 8500. Traffic Mirroring adds 54 bytes of additional headers to the original packet payload when using IPv4, and 74 bytes when using IPv6. Therefore, the maximum packet size that can be delivered to an appliance without truncation is $8500 - 54 = 8446$ when using IPv4, or $8500 - 74 = 8426$ when using IPv6.
- You can use the BytesProcessed and PacketsDropped CloudWatch metrics for VPC endpoints to monitor the volume of traffic being sent over the Gateway Load Balancer endpoint. You can also use CloudWatch metrics for Traffic Mirroring. For more information, see [Monitor mirrored traffic](#).

VXLAN encapsulation

Mirrored traffic is encapsulated in VXLAN packets and then routed to the mirror target. The security groups for a traffic mirror target must allow VXLAN traffic (UDP port 4789) from the traffic mirror source. The route table for the subnet with the traffic mirror source must have a route that sends the mirrored traffic to the traffic mirror target. The monitoring software that you run on the mirror target must be able to process encapsulated VXLAN packets.

Routing and security groups

Encapsulated mirror traffic is routed to the traffic mirror target by using the VPC route table. Make sure that your route table is configured to send the mirrored traffic to the traffic mirror target.

Inbound traffic that is dropped at the traffic mirror source, because of the inbound security group rules or the inbound network ACL rules, is not mirrored.

Mirrored outbound traffic is not subject to the outbound security group rules for the traffic mirror source.

Processing mirrored traffic

You can use open-source tools or choose a monitoring solution available on [AWS Marketplace](#). You can stream mirrored traffic to any network packet collector or analytics tool, without having to install vendor-specific agents.

Understand traffic mirror filter concepts

A *traffic mirror filter* is a set of inbound and outbound rules that determines which traffic is copied from the traffic mirror source and sent to the traffic mirror target. You can also choose to mirror certain network services traffic, including Amazon DNS. When you add network services traffic, all traffic (inbound and outbound) related to that network service is mirrored.

We evaluate traffic mirror filter rules from the lowest value to the highest value. The first rule that matches the traffic determines whether the traffic is mirrored. If you don't add any rules, then no traffic is mirrored.

For example, in the following set of filter rules, rule 10 ensures that SSH traffic from my network to my VPC is not mirrored and rule 20 mirrors all other IPv4 TCP traffic.

Number	Rule action	Protocol	Source port range	Destination port range	Source CIDR block	Destination CIDR block
10	reject	TCP (6)		22	<i>my-network</i>	<i>vpc-cidr</i>
20	accept	TCP (6)			0.0.0.0/0	0.0.0.0/0

In the following set of filter rules, rule 10 mirrors HTTPS traffic from all IPv4 addresses and rule 20 mirrors HTTPS traffic from all IPv6 addresses.

Number	Rule action	Protocol	Source port range	Destination port range	Source CIDR block	Destination CIDR block
10	accept	TCP (6)		443	0.0.0.0/0	0.0.0.0/0
20	accept	TCP (6)		443	::/0	::/0

Note that if you don't add outbound rules, then no outbound traffic is mirrored.

Understand traffic mirror session concepts

A *traffic mirror session* establishes a relationship between a traffic mirror source and a traffic mirror target. Traffic mirror sessions are evaluated based on the ascending session number that you define when you create the session.

A traffic mirror session contains the following resources:

- A traffic mirror [source](#)
- A traffic mirror [target](#)
- A traffic mirror [filter](#)

Each packet is mirrored once. However, you can use multiple traffic mirror sessions on the same mirror source. This is useful if you want to send a subset of the mirrored traffic from a traffic mirror source to multiple tools. For example, you can filter HTTP traffic in a higher priority traffic mirror session and send it to a specific monitoring appliance. At the same time, you can filter all other TCP traffic in a lower priority traffic mirror session and send it to another monitoring appliance.

Traffic mirror sources

A traffic mirror source is the network interface of type `interface`. For example, a network interface for an EC2 instance or an RDS instance.

A network interface can't be used as a traffic mirror source if the same Elastic network interface is already in use in an existing traffic mirror target.

Note

If the source network interface gets deleted, AWS deletes the traffic mirroring session.

Traffic Mirroring is not available on all instance types.

Virtualized instance types in the following instance families are supported as Traffic Mirroring source:

- **General purpose:** A1 | M4 | M5 | M5a | M5ad | M5d | M5dn | M5n | M5zn | M6a | M6g | M6gd | M6i | M6id | M6idn | M6in | M7a | M7g | M7gd | M7i | M7i-flex | Mac1 | Mac2 | Mac2-m1ultra | Mac2-m2 | Mac2-m2pro | T3 | T3a | T4g
- **Compute optimized:** C4 | C5 | C5a | C5ad | C5d | C5n | C6a | C6g | C6gd | C6gn | C6i | C6id | C6in | C7a | C7g | C7gd | C7i | C7i-flex
- **Memory optimized:** R4 | R5 | R5a | R5ad | R5b | R5d | R5dn | R5n | R6a | R6g | R6gd | R6i | R6id | R6idn | R6in | R7a | R7g | R7gd | R7i | R7iz | U-3tb1 | U-6tb1 | U-9tb1 | U-12tb1 | U-18tb1 | U-24tb1 | U7i-6tb | U7i-8tb | U7i-12tb | U7in-16tb | U7in-24tb | U7in-32tb | U7inh-32tb | X1 | X1e | X2gd | X2idn | X2iedn | X2iezn | z1d
- **Storage optimized:** D2 | D3 | D3en | H1 | I3 | I3en | I4g | I4i | I7i | Im4gn | Is4gen
- **Accelerated computing:** DL1 | DL2q | F1 | F2 | G3 | G4ad | G4dn | G5 | G5g | G6 | G6e | G6f | Gr6 | Gr6f | Inf1 | Inf2 | P3 | P3dn | P4d | P4de | P5 | P5e | Trn1 | Trn1n | VT1
- **High-performance computing:** Hpc6a | Hpc6id | Hpc7a

Understand traffic mirror source and target connectivity options

Connecting traffic mirror sources and targets in different VPCs can be useful for monitoring and analyzing network traffic across multiple environments. This allows you to centralize your network monitoring and security analysis, even if your application infrastructure is spread across different VPCs or AWS accounts.

The traffic mirror source and the traffic mirror target (monitoring appliance) can be in the same VPC or different VPCs and can be connected using the following resources:

- An intra-Region VPC peering connection.

- A transit gateway.
- A Gateway Load Balancer endpoint.

The traffic mirror target can be owned by an AWS account that is different from the traffic mirror source.

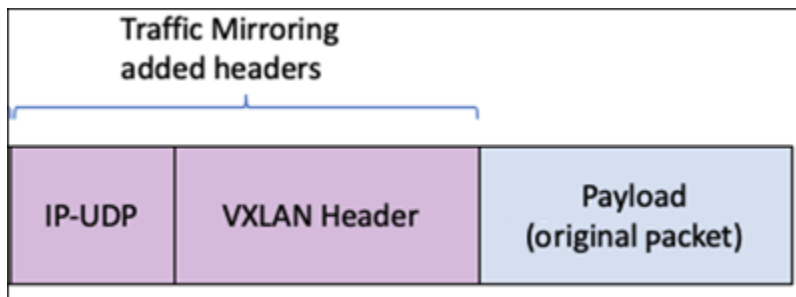
The mirrored traffic is sent to the traffic mirror target using the source VPC route table. Before you configure Traffic Mirroring, make sure that the traffic mirror source can route to the traffic mirror target.

The following table describes the available resource configurations.

Source owner	Source VPC	Target owner	Target VPC	Connectivity option
Account A	VPC 1	Account A	VPC1	No additional configuration
Account A	VPC 1	Account A	VPC 2	Intra-Region peering or a transit gateway or Gateway Load Balancer endpoint
Account A	VPC 1	Account B	VPC 2	Cross-account intra-Region peering connection, a transit gateway, or a Gateway Load Balancer endpoint
Account A	VPC 1	Account B	VPC 1	VPC sharing

Understanding traffic mirror packet format

Mirrored traffic is encapsulated with a VXLAN header. All appliances that receive traffic directly with this feature should be able parse a VXLAN-encapsulated packet, as shown in the following example:

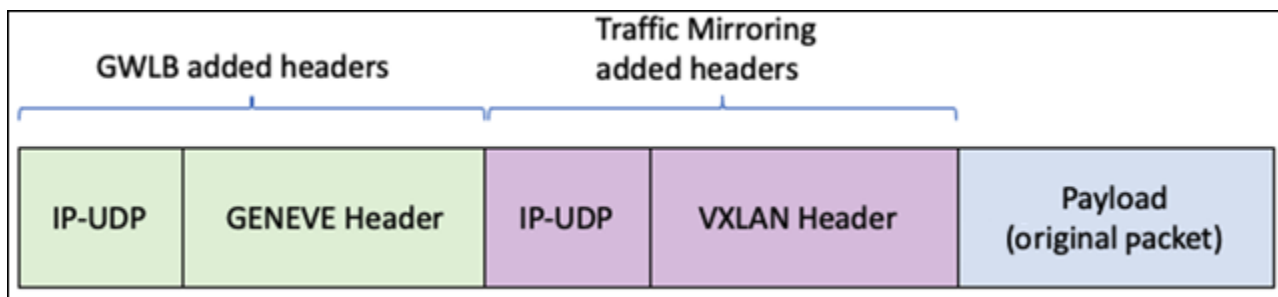


For more information about the VXLAN protocol, see [RFC 7348](#).

The following fields apply to Traffic Mirroring:

- **VXLAN ID** — The virtual network ID that you can assign to a traffic mirror session. If you do not assign a value, we assign a random value that is unique to all sessions in the account.
- **Source IP address** — The primary IP address of the source network interface.
- **Source port** — The port is determined by a 5-tuple hash of the original L2 packet, for ICMP, TCP, and UDP flows. For other flows, the port is determined by a 3-tuple hash of the original L2 packet.
- **Destination IP address** — The primary IP address of the appliance, Gateway Load Balancer endpoint, or Network Load Balancer (when the appliance is deployed behind one).
- **Destination port** — The port is 4789, which is the well known port for VXLAN.

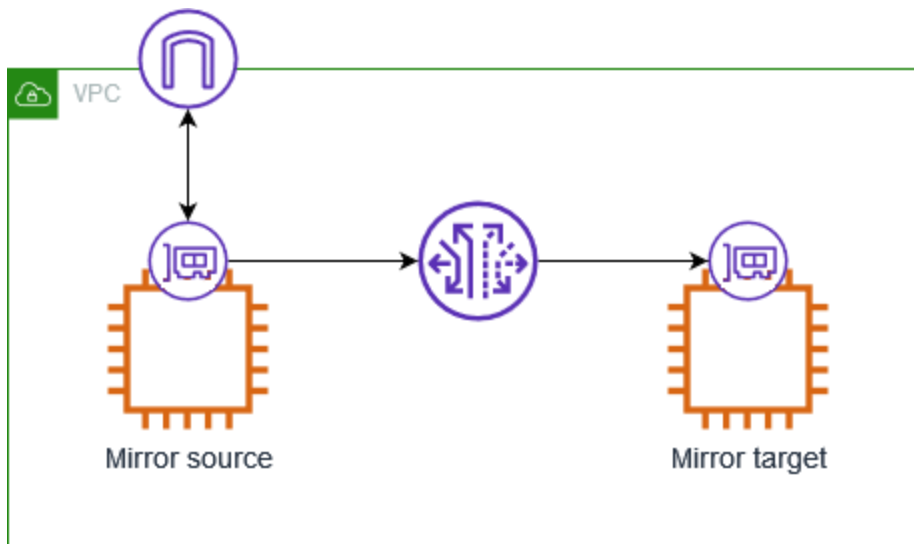
Appliances that received mirrored traffic through a Gateway Load Balancer should be able to parse both outer GENEVE encapsulation (from Gateway Load Balancer) and an inner VXLAN encapsulation (from VPC Traffic Mirroring) to retrieve the original L3 packet. The following shows an example:



Get started using Traffic Mirroring to monitor network traffic

To get started using Amazon VPC Traffic Mirroring, you'll need a VPC with at least one elastic network interface (ENI) that you want to mirror traffic from. In this section, you'll create a traffic mirror target and you'll create a traffic mirror session, which defines the parameters of the traffic mirroring operation, such as the source ENI, the mirror target, and any filtering rules. Once these components are in place, you'll start mirroring traffic to the target for monitoring and analysis.

A mirror session is a connection between a mirror source and a mirror target. In the following diagram, both the mirror source and the mirror target are EC2 instances. The mirror filter determines which network packets are mirrored. For example, you can add inbound and outbound rules to the filter such that it rejects SSH traffic but accepts all other traffic. Traffic Mirroring applies the filter rules, and then copies the accepted traffic from the network interface of the mirror source to the network interface of the mirror target. You can run your capture and analysis tools on the packets delivered to the mirror target.



Tasks

- [Prerequisites](#)
- [Step 1: Create the traffic mirror target](#)
- [Step 2: Create the traffic mirror filter](#)
- [Step 3: Create the traffic mirror session](#)
- [Step 4: Analyze the data](#)

Prerequisites

- The traffic mirror source is a network interface that you want to mirror traffic from. The traffic mirror target is the destination for the mirrored traffic, such as an EC2 instance.
- The traffic mirror source and traffic mirror target must be in the same VPC or in VPCs that are connected (for example, using VPC peering or a transit gateway).
- The traffic mirror target must allow traffic to UDP port 4789.
- The traffic mirror source must have a route table entry for the traffic mirror target.
- Security group rules and network ACL rules on the traffic mirror target cannot drop the mirrored traffic from the traffic mirror source.

Step 1: Create the traffic mirror target

Use an EC2 instance as the destination for the mirrored traffic.

To create a traffic mirror target

1. Open the Amazon VPC console at <https://console.aws.amazon.com/vpc/>.
2. In the **Region** selector, choose the AWS Region that you used when you created the EC2 instance.
3. On the navigation pane, choose **Traffic Mirroring, Mirror targets**.
4. Choose **Create traffic mirror target**.
5. (Optional) For **Name tag**, enter a name for the traffic mirror target.
6. (Optional) For **Description**, enter a description for the traffic mirror target.
7. For **Target type**, choose **Network Interface**.
8. For **Target**, choose the network interface of the EC2 instance.
9. (Optional) For each tag to add, choose **Add new tag** and enter the tag key and tag value.
10. Choose **Create**.

Step 2: Create the traffic mirror filter

A traffic mirror filter contains one or more traffic mirror rules, and a set of network services. The filters and rules that you add define the traffic that is mirrored.

To create a traffic mirror filter

1. On the navigation pane, choose **Traffic Mirroring, Mirror filters**.
2. Choose **Create traffic mirror filter**.
3. (Optional) For **Name tag**, enter a name for the traffic mirror filter.
4. (Optional) For **Description**, enter a description for the traffic mirror filter.
5. For each rule, inbound or outbound, choose **Add rule**, and then specify the following information:
 - **Number**: The rule priority.
 - **Rule action**: Indicates whether to accept or reject the packets.
 - **Protocol**: The protocol.
 - (Optional) **Source port range**: The source port range.
 - (Optional) **Destination port range**: The destination port range.
 - **Source CIDR block**: The source CIDR block.
 - **Destination CIDR block**: The destination CIDR block.
 - **Description**: A description for the rule.
6. (Optional) For each tag to add, choose **Add new tag** and enter the tag key and tag value.
7. Choose **Create**.

Step 3: Create the traffic mirror session

Create a traffic mirror session that sends mirrored packets from the source to a target so that you can monitor and analyze traffic.

To create a traffic mirror session

1. In the navigation pane, choose **Traffic Mirroring, Mirror sessions**.
2. Choose **Create traffic mirror session**.
3. (Optional) For **Name tag**, enter a name for the traffic mirror session.
4. (Optional) For **Description**, enter a description for the traffic mirror session.
5. For **Mirror source**, choose the network interface of the mirror source.
6. For **Mirror target**, choose your traffic mirror target.
7. For **Additional settings**, do the following:

- a. For **Session number**, enter **1**, which is the highest priority.
- b. (Optional) For **VNI**, enter the VXLAN ID to use for the traffic mirror session. For more information about the VXLAN protocol, see [RFC 7348](#).

If you do not enter a value, we assign a random unused number.

- c. (Optional) For **Packet length**, enter the number of bytes in each packet to mirror.

To mirror the entire packet, do not enter a value. To mirror only a portion of each packet, set this value to the number of bytes to mirror. For example, if you set this value to 100, the first 100 bytes after the VXLAN header that meet the filter criteria are copied to the target.

- d. For **Filter**, choose your traffic mirror filter.
8. (Optional) For each tag to add, choose **Add new tag** and enter the tag key and tag value.
 9. Choose **Create**.

Step 4: Analyze the data

After the mirrored traffic is copied to the traffic mirror target, you can use a tool from the [AWS Partner Network](#) to analyze the data.

Traffic Mirroring example configuration scenarios

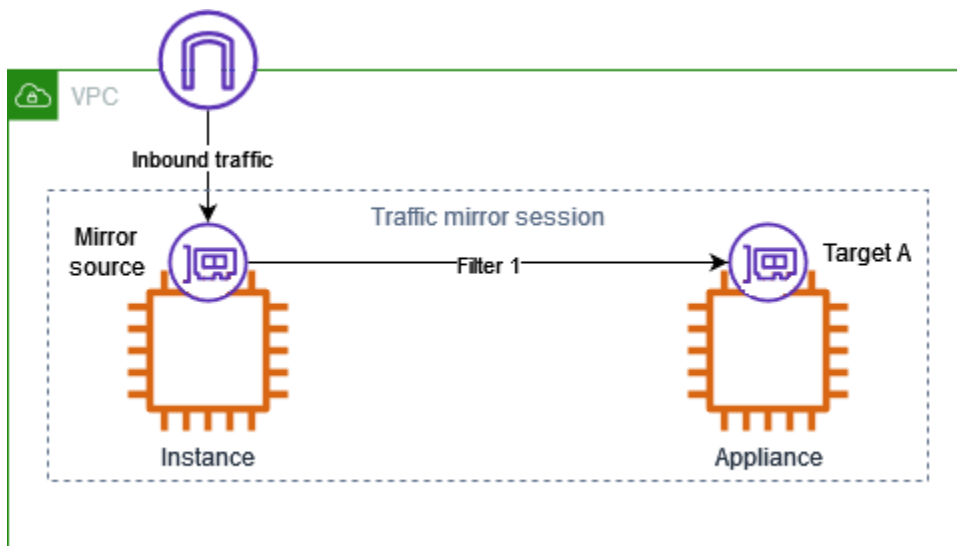
This section consists of step-by-step instructions you can use to configure Traffic Mirroring for the following scenarios:

- [Mirror inbound TCP traffic to a single appliance](#)
- [Mirror inbound TCP and UDP traffic to multiple appliances](#)
- [Mirror non-local VPC traffic](#)
- [Mirror traffic to a Gateway Load Balancer endpoint](#)

To mirror traffic from multiple network interfaces, see [VPC Traffic Mirroring Source Automation Application](#) on github.

Example: Mirror inbound TCP traffic to a single monitoring appliance

Consider the scenario where you want to mirror inbound TCP traffic on an instance, and send it to a single monitoring appliance. You need the following traffic mirror resources for this example.



Resources

- A traffic mirror target for the appliance (Target A)
- A traffic mirror filter with a traffic mirror rule for the TCP inbound traffic (Filter 1)

- A traffic mirror session that has the following:
 - A traffic mirror source
 - A traffic mirror target for the appliance
 - A traffic mirror filter with a traffic mirror rule for the TCP inbound traffic

Step 1: Create a traffic mirror target

Create a traffic mirror target (Target A) for the monitoring appliance. Depending on your configuration, the target is one of the following types:

- The network interface of the monitoring appliance
- The Network Load Balancer when the appliance is deployed behind one
- The Gateway Load Balancer endpoint when the appliance is deployed behind a Gateway Load Balancer

For more information, see [the section called "Create or delete a traffic mirror target"](#).

Step 2: Create a traffic mirror filter

Create a traffic mirror filter (Filter 1) that has the following inbound rule. For more information, see [the section called "Create, modify, or delete a traffic mirror filter"](#).

Option	Value
Rule action	Accept
Protocol	TCP
Source port range	
Destination port range	
Source CIDR block	0.0.0.0/0
Destination CIDR block	0.0.0.0/0
Description	TCP Rule

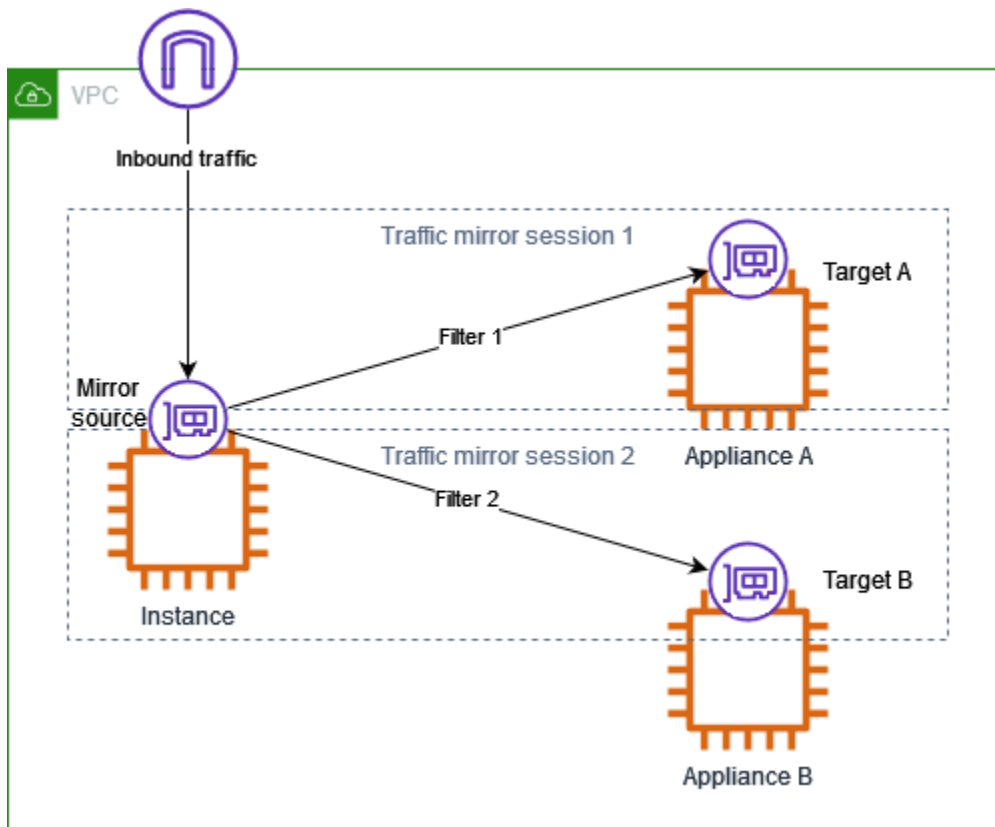
Step 3: Create a traffic mirror session

Create and configure a traffic mirror session with the following options. For more information, see [the section called “Create, modify, or delete a traffic mirror session”](#).

Option	Value
Mirror source	The network interface of the instance that you want to monitor.
Mirror target	Target A
Filter	Filter 1

Example: Mirror inbound TCP and UDP traffic to multiple appliances

Consider the scenario where you want to mirror inbound TCP and UDP traffic on an instance. But you want to send the TCP traffic to one appliance (Appliance A), and the UDP traffic to a second appliance (Appliance B). You need the following traffic mirror entities for this example.



Resources

- A traffic mirror target for Appliance A (Target A)
- A traffic mirror target for Appliance B (Target B)
- A traffic mirror filter with a traffic mirror rule for the TCP inbound traffic (Filter 1)
- A traffic mirror filter with a traffic mirror rule for the UDP inbound traffic (Filter 2)
- A traffic mirror session that has the following:
 - A traffic mirror source
 - A traffic mirror target (Target A) for Appliance A
 - A traffic mirror filter (Filter 1) with a traffic mirror rule for the TCP inbound traffic
- A traffic mirror session that has the following:
 - A traffic mirror source
 - A traffic mirror target (Target B) for Appliance B
 - A traffic mirror filter (Filter 2) with a traffic mirror rule for the UDP inbound traffic

Step 1: Create a traffic mirror target for Appliance A

Create a traffic mirror target for Appliance A (Target A). Depending on your configuration, the target is one of the following types:

- The network interface of the monitoring appliance
- The Network Load Balancer when the appliance is deployed behind one
- The Gateway Load Balancer endpoint when the appliance is deployed behind a Gateway Load Balancer

For more information, see [the section called “Create or delete a traffic mirror target”](#).

Step 2: Create a traffic mirror target for Appliance B

Create a traffic mirror target (Target B) for Appliance B. Depending on your configuration, the target is one of the following types:

- The network interface of the monitoring appliance
- The Network Load Balancer when the appliance is deployed behind one
- The Gateway Load Balancer endpoint when the appliance is deployed behind a Gateway Load Balancer

For more information, see [the section called “Create or delete a traffic mirror target”](#).

Step 3: Create a traffic mirror filter with a rule for TCP traffic

Create a traffic mirror filter (Filter 1) with the following inbound rule for TCP traffic. For more information, see [the section called “Create, modify, or delete a traffic mirror filter”](#)

Option	Value
Rule action	Accept
Protocol	TCP
Source port range	
Destination port range	

Option	Value
Source CIDR block	0.0.0.0/0
Destination CIDR block	0.0.0.0/0
Description	TCP Rule

Step 4: Create a traffic mirror filter with a rule for UDP traffic

Create a traffic mirror filter (Filter 2) with the following inbound rule for UDP traffic. For more information, see [the section called “Create, modify, or delete a traffic mirror filter”](#)

Option	Value
Rule action	Accept
Protocol	UDP
Source port range	
Destination port range	
Source CIDR block	0.0.0.0/0
Destination CIDR block	0.0.0.0/0
Description	UDP Rule

Step 5: Create a traffic mirror session for the TCP traffic

Create and configure a traffic mirror session with the following options. For more information, see [the section called “Create, modify, or delete a traffic mirror session”](#).

Option	Value
Mirror source	The network interface of the instance that you want to monitor.

Option	Value
Mirror target	Target A
Filter	Filter 1
Session number	1

Step 6: Create a traffic mirror session for the UDP traffic

Create and configure a traffic mirror session with the following options. For more information, see [the section called "Create, modify, or delete a traffic mirror session"](#).

Option	Value
Mirror source	The network interface of the instance that you want to monitor.
Mirror target	Target B
Filter	Filter 2
Session number	2

Example: Mirror non-local VPC traffic

Consider the scenario where you want to monitor traffic leaving your VPC or traffic whose source is outside your VPC. In this case, you will mirror all traffic except traffic passing within your VPC and send it to a single monitoring appliance. You need the following traffic mirror resources:

- A traffic mirror target for the appliance (Target A)
- A traffic mirror filter that has two sets of rules for outbound and inbound traffic. For outbound traffic, it will reject all packets which have a destination IP in the VPC CIDR block and accept all other outbound packets. For inbound traffic, it will reject all packets which have a source IP in the VPC CIDR block and accept all other inbound packets.
- A traffic mirror session that has the following:
 - A traffic mirror source

- A traffic mirror target for the appliance (Target A)
- A traffic mirror filter with a traffic mirror rule for the TCP inbound traffic (Filter F)

In this example, the VPC CIDR block is 10.0.0.0/16.

Step 1: Create a traffic mirror target

Create a traffic mirror target (Target A) for the monitoring appliance. Depending on your configuration, the target is one of the following types:

- The network interface of the monitoring appliance
- The Network Load Balancer when the appliance is deployed behind one
- The Gateway Load Balancer endpoint when the appliance is deployed behind a Gateway Load Balancer

For more information, see [the section called “Create or delete a traffic mirror target”](#).

Step 2: Create a traffic mirror filter

Create a traffic mirror filter (Filter F) that has the following rules. For more information, see [the section called “Create, modify, or delete a traffic mirror filter”](#).

Outbound traffic mirror filter rules

Create the following outbound rules:

- Reject all outbound packets which have a destination IP in the VPC CIDR block
- Accept all other outbound packets (destination CIDR block 0.0.0.0/0)

Option	Value
Rule number	10
Rule action	Reject
Protocol	All

Option	Value
Source port range	
Destination port range	
Source CIDR block	0.0.0.0/0
Destination CIDR block	10.0.0.0/16
Description	Reject all intra-VPC traffic

Option	Value
Rule number	20
Rule action	Accept
Protocol	All
Source port range	
Destination port range	
Source CIDR block	0.0.0.0/0
Destination CIDR block	0.0.0.0/0
Description	Accept all outbound traffic

Inbound traffic mirror filter rules

Create the following inbound rules:

- Reject all inbound packets which have a source IP in the VPC CIDR block
- Accept all other inbound packets (source CIDR block 0.0.0.0/0)

Option	Value
Rule number	10
Rule action	Reject
Protocol	All
Source port range	
Destination port range	
Source CIDR block	10.0.0.0/16
Destination CIDR block	0.0.0.0/0
Description	Reject all intra-VPC traffic

Option	Value
Rule number	20
Rule action	Accept
Protocol	All
Source port range	
Destination port range	
Source CIDR block	0.0.0.0/0
Destination CIDR block	0.0.0.0/0
Description	Accept all inbound traffic

Step 3: Create a traffic mirror session

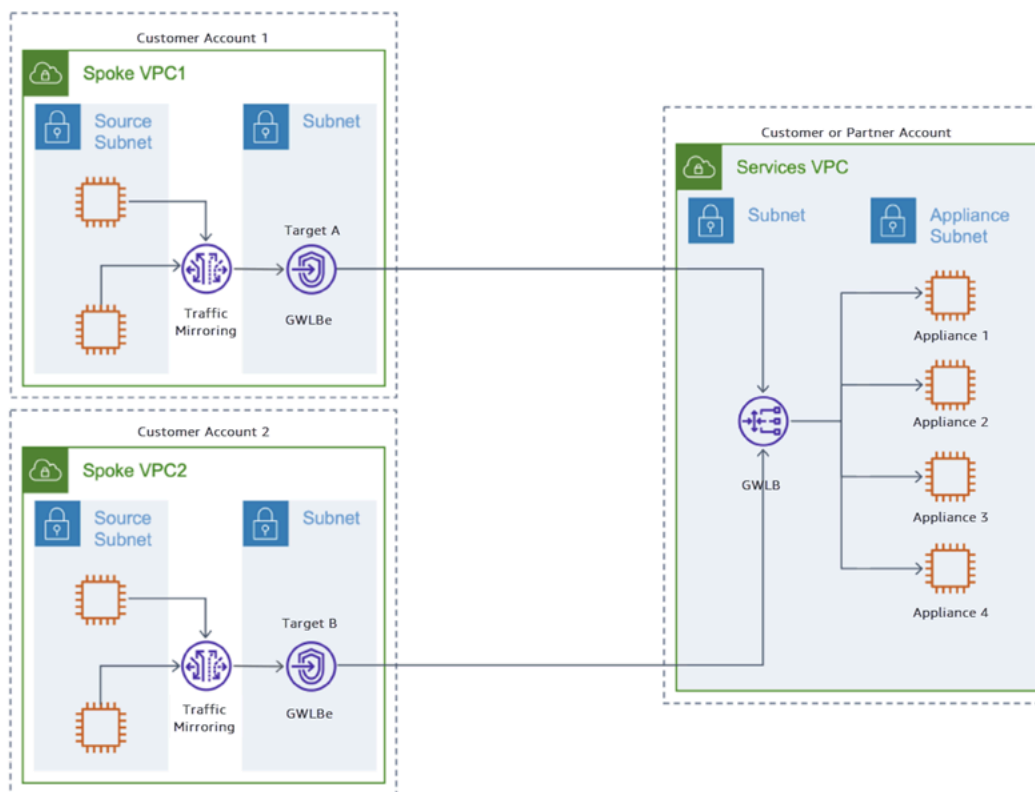
Create and configure a traffic mirror session with the following options. For more information, see [the section called “Create, modify, or delete a traffic mirror session”](#).

Option	Value
Mirror source	The network interface of the instance that you want to monitor.
Mirror target	Target A
Filter	Filter F

Example: Mirror traffic to appliances behind a Gateway Load Balancer using Gateway Load Balancer endpoints

You can deploy a Gateway Load Balancer (GWLB) and Gateway Load Balancer endpoint (GWLBe) to securely send mirror traffic across VPC and accounts. The GWLBe is a VPC endpoint that provides private connectivity between VPC with the mirror sources and the monitoring appliances deployed behind the GWLB.

The following diagram shows a deployment of a GWLB for traffic mirroring utilizing GWLBe interfaces. The GWLB is deployed in a centralized Service VPC with multiple appliances as targets. The GWLB is set up for each Availability Zone that the customer wants to monitor traffic, and it can configure their GWLB with cross-zone load balancing as an option to protect against single Availability Zone failures. In the spoke VPCs, GWLBe interfaces are deployed in each spoke VPC. These endpoints are connected to the GWLB to send traffic from the spoke VPC to the Service VPC.



Consider the scenario where you want to mirror inbound TCP traffic on an instance and then send it to a Gateway Load Balancer using a Gateway Load Balancer endpoint. You need the following Traffic Mirroring entities for this example:

- A Traffic Mirroring target for the Gateway Load Balancer endpoint (Target A) in Spoke VPC1
- A Traffic Mirroring target for the Gateway Load Balancer endpoint (Target B) in Spoke VPC2
- A Traffic Mirroring filter with a Traffic Mirroring rule for the TCP inbound traffic (Filter 1) for the Gateway Load Balancer endpoint
- A Traffic Mirroring session for Spoke VPC1 that has the following:
 - A Traffic Mirroring source
 - A Traffic Mirroring target (Target A) for the Gateway Load Balancer endpoint
 - A Traffic Mirroring filter (Filter 1) with a Traffic Mirroring rule for the TCP inbound traffic
- A Traffic Mirroring session for Spoke VPC2 that has the following:
 - A Traffic Mirroring source
 - A Traffic Mirroring target (Target B) for the Gateway Load Balancer endpoint
 - A Traffic Mirroring filter (Filter 1) with a Traffic Mirroring rule for the TCP inbound traffic

Step 1: Create a traffic mirror target in Spoke VPC1

Create a traffic mirror target (Target A) for the Gateway Load Balancer endpoint in Spoke VPC1. For more information, see [Create or delete a traffic mirror target](#).

The Gateway Load Balancer endpoint will be the target when the monitoring appliances are deployed behind a Gateway Load Balancer.

Step 2: Create a traffic mirror target in Spoke VPC2

Create a traffic mirror target (Target B) for the Gateway Load Balancer endpoint in Spoke VPC1. For more information, see [Create or delete a traffic mirror target](#).

The Gateway Load Balancer endpoint will be the target when the monitoring appliances are deployed behind a Gateway Load Balancer.

Step 3: Create a traffic mirror filter rule

Create a traffic mirror filter (Filter 1) that has the following inbound rule. For more information on creating a filter, see [Create, modify, or delete a traffic mirror filter](#).

Option	Value
Rule action	Accept
Protocol	TCP
Source port range	
Destination port range	
Source CIDR block	0.0.0.0/0
Destination CIDR block	0.0.0.0/0
Description	TCP Rule

Step 4: Create a traffic mirror session in Spoke VPC1

Create and configure a traffic mirror session with the following options. For more information, see [the section called “Create, modify, or delete a traffic mirror session”](#).

Option	Value
Mirror source	The network interface of the instance that you want to monitor.
Mirror target	Target A
Filter	Filter 1

Step 5: Create a traffic mirror session in Spoke VPC2

Create and configure a traffic mirror session with the following options. For more information, see [the section called “Create, modify, or delete a traffic mirror session”](#).

Option	Value
Mirror source	The network interface of the instance that you want to monitor.
Mirror target	Target B
Filter	Filter 1

Work with Traffic Mirroring to copy network traffic

Traffic Mirroring allows you to create, manage, and share traffic mirror targets. These targets capture and forward a copy of your traffic to the destination of your choice. Whether you're a network administrator, security analyst, or DevOps engineer, Traffic Mirroring enables you to proactively identify and address network issues, ensure compliance, and optimize your overall network performance.

With Traffic Mirroring, you can create and delete traffic mirror targets, view and modify their configurations, and even share them with other AWS accounts. You can also construct custom traffic mirror filters to capture only the data you need, and set up, modify, or delete traffic mirror sessions to control the flow of mirrored data. By leveraging these powerful features, you can unlock new insights and make informed decisions about your infrastructure, ultimately enhancing your organization's overall network visibility and security.

By using Traffic Mirroring, you can gain visibility into your network, enabling you to make more informed decisions, improve security posture, and drive greater operational efficiency across your AWS environment.

You can work with traffic mirror targets, sessions, and filters by using the Amazon VPC console or the AWS CLI.

Contents

- [Create or delete a traffic mirror target](#)
- [View traffic mirror targets and modify target tags](#)
- [Share a traffic mirror target](#)
- [Accept or delete a shared traffic mirror target](#)
- [Create, modify, or delete a traffic mirror filter](#)
- [Create, modify, or delete a traffic mirror session](#)

Create or delete a traffic mirror target

A traffic mirror target is the destination for mirrored traffic. For more information, see [the section called "Targets"](#).

Before you can delete a traffic mirror target, you must remove it from any traffic mirror sessions.

After you create a target, assign it to a traffic mirror session. For more information, see [the section called “Create, modify, or delete a traffic mirror session”](#).

You must configure a security group for the traffic mirror target that allows VXLAN traffic (UDP port 4789) from the traffic mirror source.

You can share a traffic mirror target across accounts. For more information, see [Share a traffic mirror target](#).

To create a traffic mirror target using the console

1. Open the Amazon VPC console at <https://console.aws.amazon.com/vpc/>.
2. In the **Region** selector, choose the AWS Region that you used when you created the resource to use as the destination for the mirrored traffic.
3. On the navigation pane, choose **Traffic Mirroring, Mirror targets**.
4. Choose **Create traffic mirror target**.
5. (Optional) For **Name tag**, enter a name for the traffic mirror target.
6. (Optional) For **Description**, enter a description for the traffic mirror target.
7. For **Target type**, choose the type of the traffic mirror target:
 - **Network interface**
 - **Network Load Balancer**
 - **Gateway Load Balancer endpoint**
8. For **Target**, choose the destination resource. We display resources based on the target type that you selected in the previous step.
9. (Optional) For each tag to add, choose **Add new tag** and enter the tag key and tag value.
10. Choose **Create**.

To create a traffic mirror target using the AWS CLI

Use the [create-traffic-mirror-target](#) command.

To delete a traffic mirror target using the console

1. Open the Amazon VPC console at <https://console.aws.amazon.com/vpc/>.
2. On the navigation pane, choose **Traffic Mirroring, Mirror targets**.

3. Select the traffic mirror target and choose **Delete**.
4. When prompted for confirmation, enter **delete**, and then choose **Delete**.

To delete a traffic mirror target using the AWS CLI

Use the [delete-traffic-mirror-target](#) command.

View traffic mirror targets and modify target tags

A traffic mirror target is the destination for mirrored traffic. For more information, see [the section called "Targets"](#).

Complete the steps in this section to view traffic mirror targets or modify target tags.

To view your traffic mirror targets and modify tags using the console

1. Open the Amazon VPC console at <https://console.aws.amazon.com/vpc/>.
2. On the navigation pane, choose **Traffic Mirroring, Mirror targets**.
3. To view a target, select the ID of the traffic mirror target to open its details page.
4. To modify the tags, on the **Tags** tab, choose **Manage tags**.
5. (Optional) For each tag to add, choose **Add new tag** and enter the tag key and tag value. For each tag to remove, choose **Remove**.
6. Choose **Save**.

To view your traffic mirror targets using the AWS CLI

Use the [describe-traffic-mirror-targets](#) command.

To modify your traffic mirror target tags using the AWS CLI

Use the [create-tags](#) command to add a tag. Use the [delete-tags](#) command to remove a tag.

Share a traffic mirror target

A traffic mirror target can be owned by an AWS account that is different from the traffic mirror source.

You can use AWS Resource Access Manager (RAM) to share a traffic mirror target across accounts. Use the following procedure to share a traffic mirror target that you own.

You must create a traffic mirror target before you share it. For more information, see [the section called “Create or delete a traffic mirror target”](#).

To share a traffic mirror target

1. Open the AWS Resource Access Manager console at <https://console.aws.amazon.com/ram/>.
2. Choose **Create a resource share**.
3. Under **Description**, for **Name**, enter a descriptive name for the resource share.
4. For **Select resource type**, choose **Traffic Mirror Targets**. Select the traffic mirror target.
5. For **Principals**, add principals to the resource share. For each AWS account, OU, or organization, specify its ID and choose **Add**.

For **Allow external accounts**, choose whether to allow sharing for this resource with AWS accounts that are external to your organization.

6. (Optional) Under **Tags**, enter a tag key and tag value pair for each tag. These tags are applied to the resource share but not to the traffic mirror target.
7. Choose **Create resource share**.

Accept or delete a shared traffic mirror target

Before you can use a cross-account traffic mirror target, the traffic mirror target owner shares the target with you by using the AWS Resource Access Manager. When you are in different AWS Organizations from the owner, after the owner shares the traffic mirror target, you accept the share request. After you accept the share request, you can use the traffic mirror target in a traffic mirror session.

The traffic mirror target is visible to shared accounts in their `DescribeTrafficMirrorTarget` API calls. Only the traffic mirror target owner can modify or delete the traffic mirror target.

Traffic mirror sessions that are created in a different account than the traffic mirror target are visible in `DescribeTrafficMirrorSession` API calls that are made by the traffic mirror target owner.

If you are in different AWS Organizations from the share owner, you must accept the resource share before you can access the shared resources.

You can delete a resource share at any time. When you delete a resource share, all principals that are associated with the resource share lose access to the shared resources. Deleting a resource share does not delete the shared resources.

When you delete a shared traffic mirror target that is in use, the traffic mirror session becomes inactive.

To accept or delete a shared traffic mirror target

1. Open the AWS Resource Access Manager console at <https://console.aws.amazon.com/ram/>.
2. To accept a shared traffic mirror target, in the navigation pane, choose **Shared with me, Resource shares**.
3. Select the resource share.
4. Choose **Accept resource share**.
5. To view the shared traffic mirror target, open the **Traffic Mirror Targets** page in the Amazon VPC console.
6. To delete a shared traffic mirror target, on the navigation pane, choose **Shared by me, Resource shares**.
7. Select the resource share.

Be sure to select the correct resource share. You cannot recover a resource share after you delete it.

8. Choose **Delete**.
9. When prompted for confirmation, enter **delete**, and then choose **Delete**.

Create, modify, or delete a traffic mirror filter

Use a traffic mirror filter and its rules to determine the traffic that is mirrored. A traffic mirror filter contains one or more traffic mirror rules. For more information, see [the section called "Filters"](#).

Rules are evaluated from the lowest value to the highest value. The first rule that matches the traffic determines the action to take.

Before you can delete a traffic mirror filter, you must remove it from any traffic mirror sessions.

To create, modify, or delete a traffic mirror filter using the console

1. Open the Amazon VPC console at <https://console.aws.amazon.com/vpc/>.

2. On the navigation pane, choose **Traffic Mirroring, Mirror filters**.
3. To delete a filter, select the traffic mirror filter, and then choose **Actions, Delete**.
4. When prompted for confirmation, enter **delete**, and then choose **Delete**.
5. To modify a filter, select the ID of the traffic mirror filter to open its details page. For each rule to add, choose either **Inbound rules**, **Add inbound rule** or **Outbound rules**, and then choose **Actions** and modify the rule.
6. To create a filter, choose **Create traffic mirror filter**.
7. (Optional) For **Name tag**, enter a name for the traffic mirror filter.
8. (Optional) For **Description**, enter a description for the traffic mirror filter.
9. (Optional) If you need to mirror Amazon DNS traffic, select **amazon-dns**.
10. For each rule, inbound or outbound, choose **Add rule**, and then specify the following information:
 - **Number**: The rule priority.
 - **Rule action**: Indicates whether to accept or reject the packets.
 - **Protocol**: The protocol.
 - (Optional) **Source port range**: The source port range.
 - (Optional) **Destination port range**: The destination port range.
 - **Source CIDR block**: The source CIDR block. The source and destination CIDR blocks must both be either IPv4 ranges or IPv6 ranges.
 - **Destination CIDR block**: The destination CIDR block. The source and destination CIDR blocks must both be either IPv4 ranges or IPv6 ranges.
 - **Description**: A description for the rule.
11. (Optional) For each tag to add, choose **Add new tag** and enter the tag key and tag value.
12. Choose **Create**.

To create a traffic mirror filter using the AWS CLI

Use the [create-traffic-mirror-filter](#) command.

To delete a traffic mirror filter using the AWS CLI

Use the [delete-traffic-mirror-filter](#) command.

Create, modify, or delete a traffic mirror session

A traffic mirror session establishes a relationship between a traffic mirror source and a traffic mirror target. For more information, see [the section called "Sessions"](#).

You can create a traffic mirror session only if you are the owner of the network interface or the subnet for the traffic mirror source.

To create, modify, or delete a traffic mirror session using the console

1. Open the Amazon VPC console at <https://console.aws.amazon.com/vpc/>.
2. In the **Region** selector, choose the AWS Region that you used when you created the VPCs.
3. In the navigation pane, choose **Traffic Mirroring, Mirror sessions**.
4. Select the traffic mirror session, and then choose **Actions, Delete**.
5. When prompted for confirmation, enter **delete**, and then choose **Delete**.
6. To modify a session, select the radio button for the traffic mirror session.
7. Choose **Actions, Modify session**.
8. To create a session, choose **Create traffic mirror session**.
9. (Optional) For **Name tag**, enter a name for the traffic mirror session.
10. (Optional) For **Description**, enter a description for the traffic mirror session.
11. For **Mirror source**, choose the network interface of the mirror source.
12. For **Mirror target**, choose an existing traffic mirror target or choose **Create target** to create one. For more information, see [the section called "Create or delete a traffic mirror target"](#).

If the mirror target is owned by another account and shared with you, you must first [accept the resource share](#).

13. For **Additional settings**, do the following:
 - a. For **Session number**, enter the session number. The valid values are 1 to 32,766, where 1 is the highest priority. Sessions are evaluated based on the priority indicated by this session number.
 - b. (Optional) For **VNI**, enter the VXLAN ID to use for the traffic mirror session. For more information about the VXLAN protocol, see [RFC 7348](#).

If you do not enter a value, we assign a random number.
 - c. (Optional) For **Packet Length**, enter the number of bytes in each packet to mirror.

To mirror the entire packet, do not enter a value. To mirror only a portion of each packet, set this value to the number of bytes to mirror. For example, if you set this value to 100, the first 100 bytes after the VXLAN header that meet the filter criteria are copied to the target.

- d. For **Filter**, choose an existing traffic mirror filter. Alternatively, choose **Create filter**. For more information, see [the section called “Step 2: Create the traffic mirror filter”](#).

14. (Optional) For each tag to add, choose **Add new tag** and enter the tag key and tag value.

15. If you're modifying a session, choose **Modify**. If you're creating a session, choose **Create**.

To create a traffic mirror session using the AWS CLI

Use the [create-traffic-mirror-session](#) command.

Work with open-source tools for traffic mirroring

You can use open-source tools to monitor network traffic from Amazon EC2 instances. The following tools work with Traffic Mirroring:

- **Zeek** — For more information, see the [Zeek website](#).
- **Suricata** — For more information see the [Suricata website](#).

These open-source tools support VXLAN decapsulation, and they can be used at scale to monitor VPC traffic. For information about how Zeek handles VXLAN support and to download the code, see [Zeek vxlan](#) on the GitHub website. For information about how Suricata handles VXLAN support and to download the code, see [Suricata](#) on the GitHub website.

The following example uses the Suricata open-source tool. You can follow similar steps for Zeek.

Consider the scenario where you want to mirror inbound TCP traffic on an instance and send the traffic to an instance that has the Suricata software installed. You need the following traffic mirror entities for this example:

- An EC2 instance with the Suricata software installed on it
- A traffic mirror target for the EC2 instance (Target A)
- A traffic mirror filter with a traffic mirror rule for the TCP inbound traffic (Filter rule 1)
- A traffic mirror session that has the following:
 - A traffic mirror source
 - A traffic mirror target for the appliance
 - A traffic mirror filter with a traffic mirror rule for the TCP inbound traffic

Step 1: Install the Suricata software on the EC2 instance target

Launch an EC2 instance, and then install the Suricata software on it by using the following commands.

```
# Become sudo
sudo -s
# Install epel-release
amazon-linux-extras install -y epel
```

```
# Install suricata
yum install -y suricata
# Create the default suricata rules directory
mkdir /var/lib/suricata/rules
# Add a rule to match all UDP traffic
echo 'alert udp any any -> any any (msg:"UDP traffic detected"; sid:200001; rev:1;)'
  > /var/lib/suricata/rules/suricata.rules
# Start suricata listening on eth0 in daemon mode
suricata -c /etc/suricata/suricata.yaml -k none -i eth0 -D

# Capture logs can be found in /var/log/suricata/fast.log
```

Step 2: Create a traffic mirror target

Create a traffic mirror target (Target A) for the EC2 instance. Depending on your configuration, the target is one of the following types:

- The network interface of the monitoring appliance
- The Network Load Balancer when the appliance is deployed behind one.
- The Gateway Load Balancer endpoint when the appliance is deployed behind a Gateway Load Balancer

For more information, see [the section called “Create or delete a traffic mirror target”](#).

Step 3: Create a traffic mirror filter

Create a traffic mirror filter (Filter 1) with the following inbound rule. For more information, see [the section called “Create, modify, or delete a traffic mirror filter”](#).

Option	Value
Rule action	Accept
Protocol	TCP
Source port range	
Destination port range	

Option	Value
Source CIDR block	0.0.0.0/0
Destination CIDR block	0.0.0.0/0
Description	TCP Rule

Step 4: Create a traffic mirror session

Create and configure a traffic mirror session with the following options. For more information, see [the section called “Create, modify, or delete a traffic mirror session”](#).

Option	Value
Mirror source	The network interface of the instance that you want to monitor.
Mirror target	Target A
Filter	Filter 1

Monitor mirrored traffic using Amazon CloudWatch

You can monitor your mirrored traffic using Amazon CloudWatch, which collects information from your network interface that is part of a traffic mirror session, and creates readable, near real-time metrics. You can use this information to monitor and troubleshoot Traffic Mirroring.

For more information about Amazon CloudWatch, see the [Amazon CloudWatch User Guide](#). For more information, see [CloudWatch metrics that are available for your instances](#) in *Amazon EC2 User Guide*. For more information, see [Amazon CloudWatch Pricing](#).

Traffic Mirroring metrics and dimensions

The following metrics are available for your mirrored traffic at the traffic mirror source:

Metric	Description
NetworkMirrorIn	The number of bytes received on all network interfaces by the instance that are mirrored. The number reported is the number of bytes received during the period. If you are using basic (five-minute) monitoring, you can divide this number by 300 to find Bytes/second. If you have detailed (one-minute) monitoring, divide it by 60. Units: Bytes
NetworkMirrorOut	The number of bytes sent out on all network interfaces by the instance that are mirrored. The number reported is the number of bytes sent during the period. If you are using basic (five-minute) monitoring, you can divide this number by 300 to find Bytes/second. If you have detailed (one-minute) monitoring, divide it by 60.

Metric	Description
	Units: Bytes
NetworkPacketsMirrorIn	The number of packets received on all network interfaces by the instance that are mirrored. This metric is available for basic monitoring only. Units: Count
NetworkPacketsMirrorOut	The number of packets sent out on all network interfaces by the instance that are mirrored. This metric is available for basic monitoring only. Units: Count
NetworkSkipMirrorIn	The number of bytes received, that meet the traffic mirror filter rules, that did not get mirrored because of production traffic taking priority. Units: Bytes
NetworkSkipMirrorOut	The number of bytes sent out, that meet the traffic mirror filter rules, that did not get mirrored because of production traffic taking priority. Units: Bytes
NetworkPacketsSkipMirrorIn	The number of packets received, that meet the traffic mirror filter rules, that did not get mirrored because of production traffic taking priority. This metric is available for basic monitoring only. Units: Count

Metric	Description
NetworkPacketsSkipMirrorOut	The number of packets sent out, that meet the traffic mirror filter rules, that did not get mirrored because of production traffic taking priority. This metric is available for basic monitoring only. Units: Count

To filter the metric data, use the following dimensions.

Dimension	Description
AutoScalingGroupName	This dimension filters the data you request for all instances in a specified capacity group. An Auto Scaling group is a collection of instances you define if you're using Auto Scaling. This dimension is available only for Amazon EC2 metrics when the instances are in such an Auto Scaling group. Available for instances with Detailed or Basic Monitoring enabled.
ImageId	This dimension filters the data you request for all instances running this Amazon EC2 Amazon Machine Image (AMI). Available for instances with Detailed Monitoring enabled.
InstanceId	This dimension filters the data you request for the identified instance only. This helps you pinpoint an exact instance from which to monitor data. Available for instances with Detailed or Basic Monitoring enabled.
InstanceType	This dimension filters the data you request for all instances running with this specified instance type. This helps you categorize your

Dimension	Description
	data by the type of instance running. For example, you might compare data from an m1.small instance and an m1.large instance to determine which has the better business value for your application. Available for instances with Detailed Monitoring enabled.

View Traffic Mirroring CloudWatch metrics

You can view the metrics for Traffic Mirroring as follows.

To view metrics using the CloudWatch console

Metrics are grouped first by the service namespace, and then by the various dimension combinations within each namespace.

1. Open the CloudWatch console at <https://console.aws.amazon.com/cloudwatch/>.
2. In the navigation pane, choose **Metrics**.
3. Under **All metrics**, choose the **EC2** metric namespace.
4. To view the metrics, select the metric dimension.

To view metrics using the AWS CLI

At a command prompt, use the following command to list the metrics that are available for Traffic Mirroring:

```
aws cloudwatch list-metrics --namespace "AWS/EC2"
```

The Traffic Mirroring metrics are included with the metrics for Amazon EC2.

Traffic Mirroring limitations

This section contains the limitations for Traffic Mirroring.

Contents

- [General limitations](#)
- [MTU and packet length limitations](#)
- [Traffic bandwidth and prioritization limitations](#)
- [Checksum offloading limitations](#)

General limitations

This section contains general Traffic Mirroring limitations.

IPv6 traffic

Traffic Mirroring is not supported for IPv6-only subnets.

Traffic types

Traffic Mirroring can't mirror the following traffic types:

- ARP
- DHCP
- Instance metadata service
- NTP
- Windows activation

VPC Flow Logs

VPC Flow Logs do not capture mirrored traffic.

Shared VPCs and subnets

- Participants cannot describe, create, modify, or delete a traffic mirror session or target that belongs to the VPC owner. Participants can describe, create, modify, and delete a traffic mirror session or target that belongs to them.

- VPC owners cannot describe, create, modify, or delete a traffic mirror session or target that belongs to the participant.

For more information see, [Share your VPC with other accounts](#) in the *Amazon VPC User Guide*.

Sources

You can only use [requester-managed network interfaces](#) created by Amazon RDS or Amazon ElastiCache as a Traffic Mirroring source in a session.

MTU and packet length limitations

We truncate the packet to the MTU value when both of the following are true:

- The traffic mirror target is a standalone instance.
- The mirrored traffic packet size is greater than the traffic mirror target MTU value.

For example, if an 8996 byte packet is mirrored, and the traffic mirror target MTU value is 9001 bytes, the mirror encapsulation results in the mirrored packet being greater than the MTU value. In this case, the mirror packet is truncated. To prevent mirror packets from being truncated, set the traffic mirror source interface MTU value to 54 bytes less than the traffic mirror target MTU value for IPv4 and 74 bytes less than the traffic mirror target MTU value when you use IPv6. Therefore, the maximum MTU value supported by Traffic Mirroring with no packet truncation is 8947 bytes.

In addition, the packet length cannot be less than 35 bytes (for IPv4 traffic) and 55 bytes (for IPv6 traffic).

For more information about configuring the network MTU value, see [Network maximum transmission unit \(MTU\)](#) in the *Amazon EC2 User Guide*.

Traffic bandwidth and prioritization limitations

Mirrored traffic counts toward instance bandwidth. For example, if you mirror a network interface that has 1 Gbps of inbound traffic and 1 Gbps of outbound traffic, the instance must handle 4 Gbps of traffic (1 Gbps inbound, 1 Gbps mirrored inbound, 1 Gbps outbound, and 1 Gbps mirrored outbound) and your packet size should be equal to or greater than 1500 Bytes. Note that [the per flow limit for EC2 instances not in placement groups is 5 Gbps](#). For instances not in placement groups, the per flow throughput should be lower than 2.5 Gbps or mirrored packets is dropped.

By default, each Gateway Load Balancer endpoint can support a bandwidth of up to 10 Gbps per Availability Zone and automatically scales up to 100 Gbps. For more information, see [AWS PrivateLink quotas](#) in the *AWS PrivateLink Guide*.

If the network traffic exceeds the bandwidth or packet-per-second (PPS) limits of an instance, mirrored traffic is dropped. This gives production traffic priority when there is traffic congestion. However, if production traffic continues to exceed the bandwidth or PPS limits, it is also dropped. Mirrored traffic might also be dropped at lower bandwidths if the average packet size of your traffic is small.

Checksum offloading limitations

The Elastic Network Adapter (ENA) provides checksum offloading capabilities. If a packet is truncated, this might result in the packet checksum not being calculated for the mirrored packet. The following checksums are not calculated when the mirrored packet is truncated:

- If the mirror packet is truncated, the mirror packet L4 checksum is not calculated.
- If any part of the L3 header is truncated, the L3 checksum is not calculated.

If this causes issues, you can disable ENA checksum offloading on the ENA for the source. For example, use the following commands on Amazon Linux 2:

```
[ec2-user ~]$ sudo ethtool --offload eth0 tx off
[ec2-user ~]$ sudo ethtool --show-offload eth0
Features for eth0:
rx-checksumming: on
tx-checksumming: off
    tx-checksum-ipv4: off
    tx-checksum-ip-generic: off [fixed]
    tx-checksum-ipv6: off [fixed]
    tx-checksum-fcoe-crc: off [fixed]
    tx-checksum-sctp: off [fixed]
```

Traffic Mirroring quotas

The following are the quotas for Traffic Mirroring for your AWS account.

Contents

- [Sessions](#)
- [Targets](#)
- [Filters](#)
- [Throughput](#)
- [Packets](#)
- [Sources](#)

Sessions

Name	Default	Adjustable
Maximum number of sessions per account	10,000	No
Maximum number of sessions per source network interface	3	No
Maximum number of sessions for a single Gateway Load Balancer endpoint	Unlimited	Not applicable

Targets

Name	Default	Adjustable
Maximum number of targets per account	10,000	No

Filters

Name	Default	Adjustable
Maximum number of filters per account	10,000	No
Maximum number of sessions per source network interface	3	No
Maximum number of filter rules per filter	10	No

Throughput

Name	Default	Adjustable
Maximum throughput through a single Gateway Load Balancer endpoint	100 Gbps	No

Packets

Name	Default	Adjustable
Maximum number of MTUs for a Gateway Load Balancer endpoint	8,500	No

Sources

Name	Default	Adjustable
Maximum number of sources per Network Load Balancer	No limit	No
Maximum number of sources per Gateway Load Balancer endpoint	No limit	No

Name	Default	Adjustable
Maximum number of sources per target	100 or 10 depending on the instance type. For more information, see the list below ¹	No

¹ The following instance types support up to 100 sources per target. All other instance types support up to 10 sources per target.

- **General Purpose:** m5.24xlarge | m5d.24xlarge | m5dn.24xlarge | m5n.24xlarge | m5zn.12xlarge | m6a.48xlarge | m6g.16xlarge | m6gd.16xlarge | m6i.32xlarge | m6id.32xlarge | m6idn.32xlarge | m6in.32xlarge | m7a.48xlarge | m7i.48xlarge | m8a.48xlarge | m8g.48xlarge | m8gd.48xlarge | m8i.96xlarge
- **Compute Optimized:** c5.18xlarge | c5.24xlarge | c5d.18xlarge | c5d.24xlarge | c5n.18xlarge | c6a.48xlarge | c6g.16xlarge | c6gd.16xlarge | c6gn.16xlarge | c6i.32xlarge | c6id.32xlarge | c6in.32xlarge | c7a.48xlarge | c7gn.16xlarge | c7i.48xlarge | c8g.48xlarge | c8gd.48xlarge | c8gn.24xlarge | c8gn.48xlarge | c8i.96xlarge
- **Memory Optimized:** r5.24xlarge | r5b.24xlarge | r5d.24xlarge | r5dn.24xlarge | r5n.24xlarge | r6a.48xlarge | r6g.16xlarge | r6gd.16xlarge | r6i.32xlarge | r6id.32xlarge | r6idn.32xlarge | r6in.32xlarge | r7a.48xlarge | r7i.48xlarge | r7iz.32xlarge | r8a.48xlarge | r8g.48xlarge | r8gb.24xlarge | r8gd.48xlarge | r8gn.24xlarge | r8gn.48xlarge | r8i.96xlarge | u-6tb1.56xlarge | u-6tb1.112xlarge | u-9tb1.112xlarge | u-12tb1.112xlarge | u-18tb1.112xlarge | u-24tb1.112xlarge | u7i-6tb.112xlarge | u7i-8tb.112xlarge | u7i-12tb.224xlarge | u7in-16tb.224xlarge | u7in-24tb.224xlarge | u7in-32tb.224xlarge | u7inh-32tb.480xlarge | x2gd.16xlarge | x2idn.32xlarge | x2iedn.32xlarge | x2iezn.12xlarge | x8g.48xlarge | z1d.12xlarge
- **Storage Optimized:** i3en.24xlarge | i4g.16xlarge | i4i.32xlarge | i7i.48xlarge | i7ie.48xlarge | i8g.48xlarge | i8ge.48xlarge | im4gn.16xlarge
- **Accelerated Computing:** dl1.24xlarge | dl2q.24xlarge | f2.48xlarge | g5.48xlarge | g5g.16xlarge | g6.48xlarge | g6e.12xlarge | g6e.24xlarge | g6e.48xlarge | inf1.24xlarge | inf2.48xlarge | p3dn.24xlarge | p4d.24xlarge | p4de.24xlarge | p5.4xlarge | p5.48xlarge | p5e.48xlarge | p5en.48xlarge | p6-b200.48xlarge | p6e-gb200.36xlarge | trn1.32xlarge | trn1n.32xlarge | trn2.48xlarge | trn2u.48xlarge | vt1.24xlarge

- **High Performance Computing:** hpc6a.48xlarge | hpc6id.32xlarge | hpc7a.12xlarge | hpc7a.24xlarge | hpc7a.48xlarge | hpc7a.96xlarge | hpc7g.4xlarge | hpc7g.8xlarge | hpc7g.16xlarge

Identity and access management for Traffic Mirroring

AWS Identity and Access Management (IAM) is an AWS service that helps an administrator securely control access to AWS resources. Administrators control who can be *authenticated* (signed in) and *authorized* (have permissions) to use traffic mirror resources.

To allow access to traffic mirror resources, you create and attach an IAM policy to an IAM role and users or groups assume that role.

The IAM role must be given permission to use specific traffic mirror resources and API actions. When you attach a policy to a role, it allows or denies permission to perform the specified tasks on the specified resources.

You can also use resource-level permissions to restrict what resources users can use when they invoke APIs.

Example Example: CreateTrafficMirrorSession policy

The following IAM policy allows users to use the CreateTrafficMirrorSession API, but restricts the action to a specific traffic mirror target (tmt-12345645678). To create a traffic mirror session, users must also have permission to use the traffic mirror filter and network interface resources. Therefore, you must include these resources in the IAM policy attached to the role.

JSON

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": "ec2:CreateTrafficMirrorSession",
      "Resource": [
        "arn:aws:ec2:*:*:traffic-mirror-target/tmt-12345645678",
        "arn:aws:ec2:*:*:traffic-mirror-filter/*",
        "arn:aws:ec2:*:*:network-interface/*"
      ]
    }
  ]
}
```

For more information about supported traffic mirror actions, resources, and condition keys, see [Actions, resources, and condition keys for Amazon EC2](#) in the *Service Authorization Reference*.

Document history for Traffic Mirroring

The following table describes the releases for Traffic Mirroring.

Change	Description	Date
Supported instance types	Additional instance types available as Traffic Mirroring source.	August 28, 2025
Support for Gateway Load Balancer endpoints as traffic mirror targets	Send mirrored traffic to monitoring appliances registered with a Gateway Load Balancer.	May 12, 2022
Support for non-Nitro instance types	Enable Traffic Mirroring on the following non-Nitro instance types: C4, D2, G3, G3s, H1, I3, M4, P2, P3, R4, X1 and X1e.	February 10, 2021
Support for Amazon CloudWatch	Monitor your mirrored traffic using Amazon CloudWatch.	November 25, 2019
Initial release	This release introduces Traffic Mirroring.	June 25, 2019